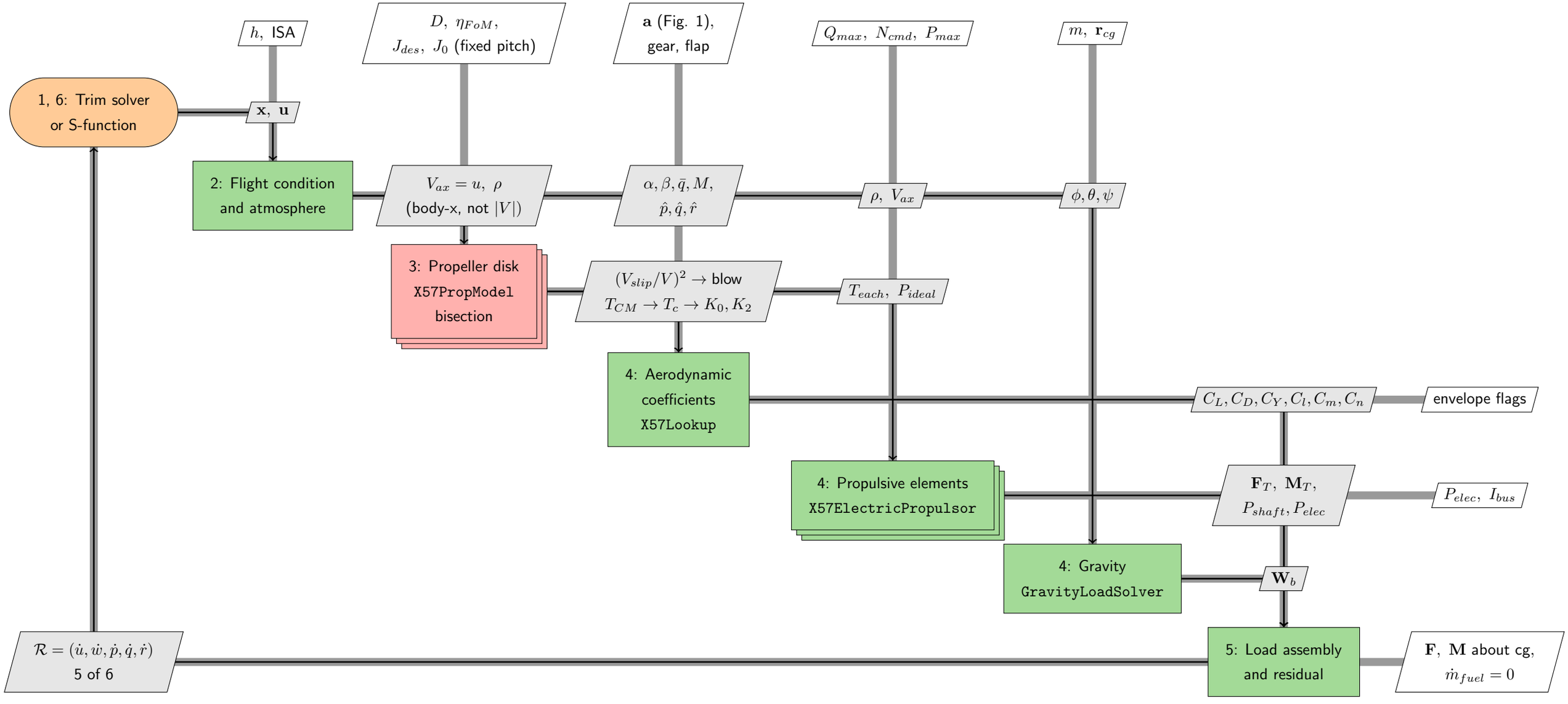

1, 6: Trim solver or S-function
h, ISA
$\mathbf{x}$, $\mathbf{u}$
D, η_{FoM}, J_{des}, J_0 (fixed pitch)
$\mathbf{a}$ (Fig. 1), gear, flap
Q_{max}, N_{cmd}, P_{max}
m, $\mathbf{r}_{cg}$
2: Flight condition and atmosphere
$V_{ax} = u$, ρ (body-x, not $|V|$)
$\alpha, \beta, \bar{q}, M$, $\hat{p}, \hat{q}, \hat{r}$
ρ, V_{ax}
ϕ, θ, ψ
3: Propeller disk X57PropModel bisection
$(V_{slip}/V)^2 \rightarrow$ blow $T_{CM} \rightarrow T_c \rightarrow K_0, K_2$
T_{each}, P_{ideal}
4: Aerodynamic coefficients X57Lookup
$C_L, C_D, C_Y, C_l, C_m, C_n$
envelope flags
4: Propulsive elements X57ElectricPropulsor
$\mathbf{F}_T$, $\mathbf{M}_T$, P_{shaft}, P_{elec}
P_{elec}, I_{bus}
4: Gravity GravityLoadSolver
$\mathbf{W}_b$
$\mathcal{R} = (\dot{u}, \dot{w}, \dot{p}, \dot{q}, \dot{r})$ 5 of 6
5: Load assembly and residual
$\mathbf{F}$, $\mathbf{M}$ about cg, $\dot{m}_{fuel} = 0$